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Forging furnace with thermochemical waste-heat recuperation by natural gas reforming: Fuel saving and heat balance

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HIGHLIGHTS

- Fuel saving by thermochemical waste-heat recuperation (TCR).
- TCR is capable of reducing fuel consumption by nearly 25%.
- TCR can be considered as hydrogen-rich fuel production technology.

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ABSTRACT

This paper considers thermochemical recuperation (TCR) of waste-heat using natural gas reforming by steam and combustion products. Combustion products contain steam (H_2O), carbon dioxide (CO_2), and ballast nitrogen (N_2). Because endothermic chemical reactions take place, methane steam-dry reforming creates new synthetic fuel that contains valuable combustion components: hydrogen (H_2), carbon monoxide (CO), and unreformed methane (CH_4). There are several advantages to performing TCR in the industrial furnaces: high energy efficiency, high regeneration rate (rate of waste-heat recovery), and low emission of greenhouse gases (CO_2, NO_x). As will be shown, the use of TCR is significantly increasing the efficiency of industrial furnaces – it has been observed that TCR is capable of reducing fuel consumption by nearly 25%. Additionally, increased energy efficiency has a beneficial effect on the environment as it leads to a reduction in greenhouse gas emissions.

Introduction

Experts from the International Energy Agency predict that oil and natural gas will be the primary fuel for the World Economy during the next 100–150 years [1]. Therefore, problems of improving energy efficiency get more relevant for many energy plants that use natural gas.

Some energy plants of the natural gas-fired power plants have almost maximum energy efficiency ($\eta = 0.9 \ldots 0.98$). However, many industrial furnaces (glass-melting furnaces, forging furnaces) have very low energy efficiency ($\eta = 0.2 \ldots 0.5$) [2]. This occurs because the industrial furnaces have a lot of waste flue heat losses. Table 1 shows the heat characteristics of various industrial furnaces. Waste flue heat loss (%) (Table 1) is the relation of enthalpy of combustion products in

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Table 1 – The heat characteristics of some industrial furnaces [1].

Furnace type	Temperature in the smelting chamber,°C	Flue gas temperature, °C	Waste flue heat loss, %
Steel-making furnace	1650–1750	1550–1600	65–75
Pit furnace	1350–1450	1250–1350	55–60
Forging furnace	1300–1400	1100–1200	55–65
Continuous furnace	1300–1400	900–1100	30–45
Glass-melting furnace	1600–1700	1000–1100	50–60

the processing chamber to the enthalpy of flue gases after the processing chamber.

To increase the energy efficiency of industrial furnaces, the heat from the waste flue gases must be recovered. In conventional industrial furnaces in order to recover the loss waste flue gases heat are used the special heat exchangers – regenerators and recuperators. This leads to a significant increase in energy efficiency in industrial furnaces. In heat exchangers occur heating of air for combustion using cooling flue gases. For example, if the air is preheated to 300–350 °C by flue gases, then the fuel economy for this industrial furnace (t_{fg} = 1200 °C, fuel type: natural gas) will be near 20–25% (by using eq. (1)). Such a waste-heat recovery type is named a thermal regeneration.

$$P = r\frac{H_2}{H_1 - H_2(1-r)}. \quad (1)$$

Here, P is the economy of fuel (natural gas) in a furnace with thermal regeneration against without thermal regeneration, %; H_1 is the enthalpy of flue gases in the processing chamber of a furnace, kJ/m^3; H_2 is the enthalpy of exhaust flue gases after regeneration system of the furnace, kJ/m^3; r is regeneration rate, [–].

In recent years, many works on thermochemical recuperation in various fuel-consuming equipment have been published for: industrial furnaces [3–5], gas turbine units [6], internal combustion engines [7], etc. Any organic fuel, the oxidation of which gives an endothermic effect, can be used for thermochemical recuperation [8,9]. The main criteria for the choice of the endothermic process for TCR are presented in the author's previous works [10]. According to the literature review, the strong interest of researchers is directed to the use of natural gas, liquid hydrocarbons, and alcohols as fuel for TCR.

A series of articles on thermochemical recuperation by methane reforming was published by Gaber and co-authors [11]. In these articles, the authors reported the benefits of using the TCR systems by natural gas reforming such as energy efficiency and the environmental performance of fuel-consuming equipment. Chakravarthy et al. presented the study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines (ICE) [12]. The results of second law analysis for various liquid fuels such as iso-octane, ethanol, and methanol were presented. The authors reported that the use of TCR for ICE leads to a considerable increase in second law efficiency for all investigated fuels: 3–5% for methanol, 9% for ethanol, 11% for iso-octane. Moreover, they noted that reduced combustion irreversibility when synfuel is used as fuel in ICE can be considered as benefits for future TCR implementations.

Kobayashi patented the thermochemical regenerative heat recovery process for industrial furnaces [13]. An industrial furnace according to his patent have at least two regenerator beds for waste-heat recuperation. While a first bed is being heated by hot flue gases produced by fuel combustion in the furnace, a second bed, heated during a previous cycle, is cooled through carrying out an endothermic chemical reforming reaction (for example, steam methane or dry methane reforming). Once the second bed is cooled by the endothermic reaction, the hot flue gases are redirected to it while the first bed, now hot, is used for carrying out the endothermic chemical reaction. Thereafter the cycle is repeated. Moreover, Barati et al. [14] investigated the thermochemical recuperation systems for recuperation thermal energy of the metallurgical slag.

Rue D. et al. [15] determined the energy efficiency of a glass furnace with thermochemical waste-heat recuperation. The authors presented the model of thermochemical waste-heat recuperation and tested it. The result of their work is the fact that energy efficiency of furnace with TCR is significantly higher than it is with thermal recuperators or other exchangers.

Thermal regeneration has some disadvantages. Heat exchangers for the preheat combustion air have a large size, mainly due to their low heat-transfer coefficient. Preheating of natural gas in the heat exchangers by means of flue gases waste heat is not widely used because it leads to high temperature with methane cracking [16]. Methane cracking results in a reduction of the heat-transfer coefficient in the exchangers because of carbon forms on the wall of the exchanger.

This disadvantage of thermal regeneration can be countered by the use of thermochemical heat regeneration of waste flue gases [17]. Thermochemical regeneration (TCR) is a prospective way to increase the energy efficiency of industrial furnaces as well as many other energetic plants that use natural gas as fuel and that have a lot of waste flue heat losses because it gives the high regeneration rate.

Thermochemical heat regeneration concepts

The basic description of TCR is the transformation of enthalpy waste-heat into the chemical energy of a new synthetic fuel that has a higher low heat value (LHV).

If in the conventional industrial furnaces conversion of chemical energy of fuel into heat energy occurs by fuel combustion, then in the industrial furnaces with TCR the conversion of chemical energy of the fuel is divided into two stages. The first stage is the increase of the LHV of the initial natural gas by transforming the enthalpy waste-heat into the chemical energy of new synthetic fuel. The second

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stage is the combustion of the new synthetic fuel that has LHV greater than the LHV of initial natural gas.

TCR by steam methane reforming (SMR)

Thermochemical heat regeneration of waste flue gases occurs when chemical endothermic reactions are used, for example, steam methane reforming, dry methane reforming, and methane cracking, etc. The idea of improving the efficiency of an energy plant by using chemical recuperation of waste heat was discussed as early as in 1972 [18].

Steam methane reforming (SMR) is widely used in the chemical industry [19–21]. It is used for hydrogen production, production of synthesis-gas, etc. The chemical reactions take the following form [22–24]:

$$CH_4 + H_2O \leftrightarrow CO + 3H_2 - 206.1 \text{ kJ/mol} \tag{2}$$

$$CO + H_2O \leftrightarrow CO_2 + H_2 + 41.1 \text{ kJ/mol} \tag{3}$$

The mixture of methane and steam that results in chemical reactions (2) and (3) produces synthesis-gas (a mixture of carbon monoxide and hydrogen) – a new synthetic fuel that has LHV greater than that of methane.

Fig. 1 shows the schematic diagram of an industrial furnace that uses TCR of waste flue gases using steam methane reforming.

The flue gases (fg) are divided into two streams after exiting the industrial furnace (IF). The first stream is passed through an air heater (AH) in order to preheat cold air (ca) creating hot air (ha). The second stream is passed through a thermochemical regenerator (TR) to heat the reaction space of the TR. Steam (s) and natural gas (ng) are fed to the TR. In the TR steam methane reforming take place by using heat from the second stream of flue gases. As a result of the reaction of SMR, a new synthetic fuel (sf) (a mixture of CO and H2) is produced. Next, the synthetic fuel (sf) and the hot air (ha) are burned in the industrial furnace.

The main advantage of this TCR process is the achievement of an almost full waste-heat recovery of flue gases keeping the components of the combustion at a relatively moderate temperature. In contrast, thermal regeneration requires very high air temperatures to achieve an almost full waste-heat recovery. The high regeneration rate can be achieved by using this way of waste-heat regeneration.

However, there are disadvantages to carrying out TCR by SMR – the biggest of which is the necessity of steam production for methane reforming. In this case, heat consumption for steam production consumes 8–10% of the power of the industrial furnace, greatly reducing the energy efficiency of the furnace. This disadvantage can be countered through the application of thermochemical waste-heat regeneration to reform methane using flue gases that contain oxidizers such as steam (H_2O) and carbon dioxide (CO_2).

TCR by methane reforming with flue gases

The TCR by methane reforming with flue gases is a promising way of increasing energy efficiency for industrial furnaces and was developed by the author of this paper. If instead of steam flue gases are used in the TCR then an increase of energy efficiency will occur, because no consumption of heat for the oxidizer production is needed.

The flue gases have a high temperature after the industrial furnace (Table 1). The flue gases generally consist of steam (H_2O), carbon dioxide (CO_2), and nitrogen (N_2). If the methane is burned, then flue gases consist of H_2O:CO_2:N_2 in an analogy of 2:1:7.52. In this case of methane reformation by TCR using flue gases, flue gases contain the steam and carbon dioxide that is used in this process. This engineering solution eliminates both the loss of heat due to the production of the oxidizer (steam) and steam preheat to high temperatures because are used flue gases with high temperatures.

Fig. 2 shows the schematic diagram of an industrial furnace that uses TCR by methane reforming with flue gases.

In this diagram, the flue gases (fg) are divided into two streams after exiting the industrial furnace (IF). The first

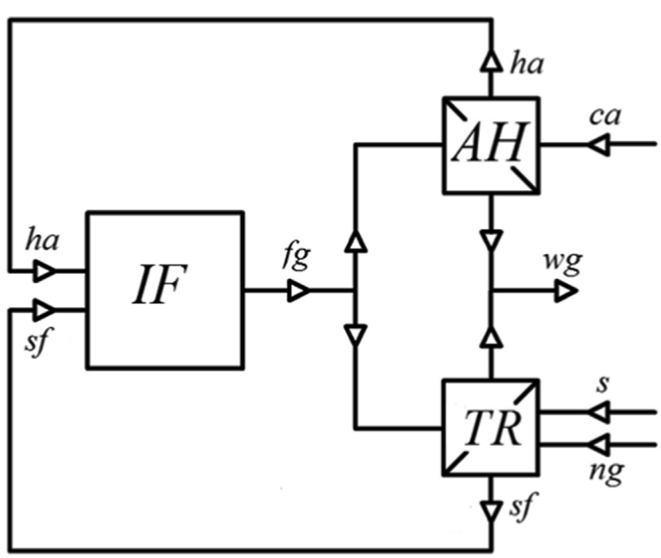

Fig. 1 – The schematic diagram of the industrial furnace with the TCR by SMR.

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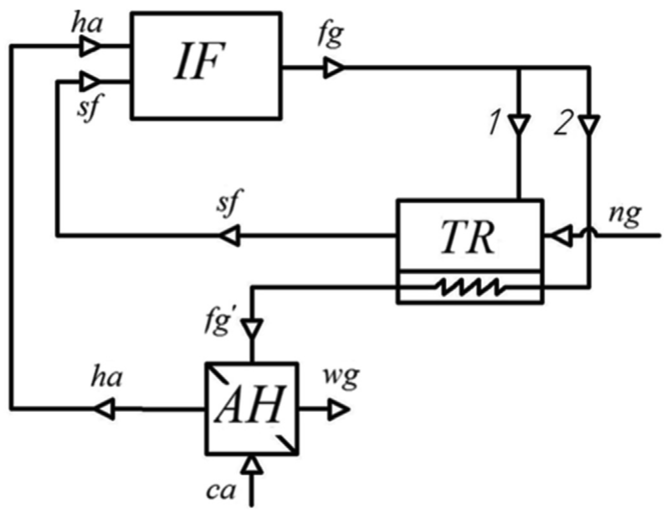

Fig. 2 – The schematic diagram of the industrial furnace with the TCR by methane reforming with flue gases.

stream (1) is fed to a reaction space of the thermochemical regenerator (TR). Also, natural gas (ng) is sent into the TR. Nickel catalysts activate the reaction space of the TR. Steam and dry reforming of the natural gas, along with other chemical reactions, take place within this reaction space. New synthetic fuel (sf) (the product gas) travels from the reaction space of the TR to the processing chamber of the industrial furnace. Concurrently, the second stream (2) from the industrial furnace is passed through a heat-exchange space of the TR as well as the air heater (AH). After exiting the air heater, the waste gases (wg) are ejected into the atmosphere like in the furnaces with thermal regeneration. Stream (2) is passed through the heat-exchange space of the TR to supply heat to induce the endothermic reaction of steam and dry reforming. The stream (fg') is also passed through the air heater to preheat the cold air (ca).

The thermochemical regenerator is the basic element of thermochemical heat regeneration. A transformation of the enthalpy of the flue gases to the chemical energy of the new synthetic fuel occurs within the thermochemical regenerator. Various catalysts accelerate the reaction of steam and dry reforming of the natural gas by activating the reaction space of the thermochemical regenerator. These catalysts are based on nickel (Ni), platinum (Pt), iridium (Ir), etc. [25].

TCR by methane reforming using flue gases opens a very big opportunity to improve the energy efficiency of industrial furnaces. This method of increasing efficiency may be applied to a wide variety of energetic plants in which are natural gas is consumed.

Thermodynamic aspects

The chemical reaction of steam and dry reforming methane occurs within the thermochemical regenerator shown in Fig. 1, while different chemical reactions occur within the regenerator. The main reactions that take place in the thermochemical regenerator are as follows [26–29]:

$$CH_4 + H_2O \leftrightarrow CO + 3H_2 - 206.1\ \text{kJ/mol} \tag{4}$$

$$CH_4 + CO_2 \leftrightarrow 2CO + 2H_2 - 247.3\ \text{kJ/mol} \tag{5}$$

$$CO + H_2O \leftrightarrow CO_2 + H_2 + 41.1\ \text{kJ/mol} \tag{6}$$

Overall reaction reforming of methane with flue gases is as follows:

$$CH_4 + k(CO_2 + 2H_2O + 7.52N_2) \leftrightarrow$$

$$\leftrightarrow aCO + bH_2 + cN_2 + dCO_2 + eH_2O + (1 - x_{CH4})CH_4 + \Delta Q_{chem} \tag{7}$$

For the stoichiometric reagent consumption the coefficients will be: $k = 1/3$, $a = 4/3$, $b = 8/3$, $c = 2.51$, $d = 0$, $e = 0$, $x_{CH4} = 1$, and $\Delta Q_{chem} = -220.3$ kJ/mol. Here, ΔQ_{chem} characterizes the thermal effect of the general reaction (7). ΔQ_{chem} also shows how the enthalpy of the flue gases has been transformed into chemical energy of the new synthetic fuel. It is obvious that the thermal effect of the overall reaction (7) depends on the conversion degree of methane:

$$\Delta Q_{chem} = x_{CH4} \cdot Q \tag{8}$$

where Q characterizes the thermal effect of the overall reaction (7) at stoichiometric reagent consumption, [kJ/mol] and x_{CH4} represents the conversion degree of methane. Thus, ΔQ_{chem} is directly proportional to the degree of conversion, x_{CH4}.

Using chemical equations (4)–(7), the schematic diagram can be presented as shown on Fig. 3.

Fig. 4 shows variations of conversion degree of methane against temperature (4a) and β ratio (4b). The β ratio characterizes the composition of the inlet mixture [30–32]:

$$\beta = \frac{(\text{inlet } H_2O + \text{inlet } CO_2)}{\text{inlet } CH_4}. \tag{9}$$

Fig. 4a shows that at equilibrium the conversion degree

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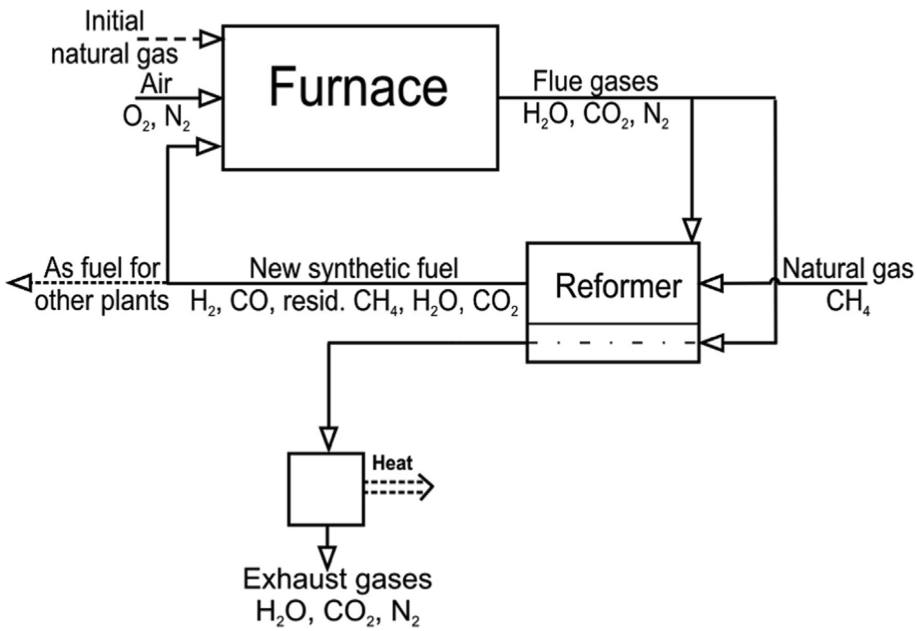

Fig. 3 – The schematic diagram of conceptual idea of the TCR for the industrial furnaces.

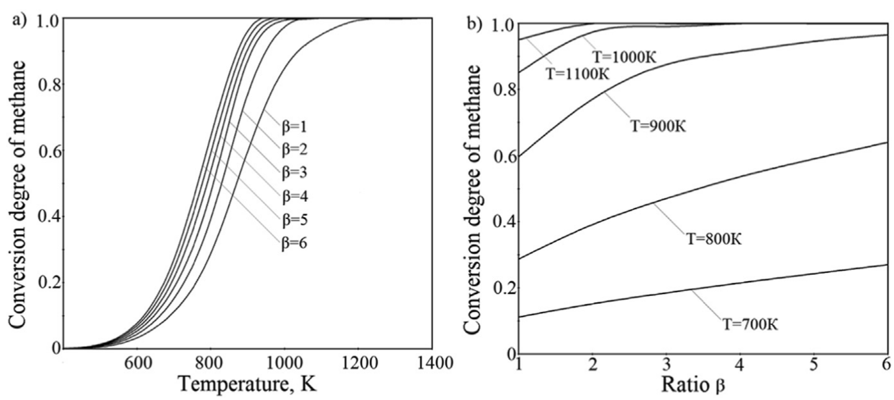

Fig. 4 – The variations of conversion degree of methane against temperature (a) in reaction space of TR and ratio β (b) at p = 1 bar.

of methane increases with increasing temperature. The optimal temperature at which TCR can be used effectively is the interval of temperature Tε(900K; 1300K). If the temperature falls below T = 900K, the conversion degree is very small. And if T > 1300K, structural damage of the thermochemical regenerator may occur. Fig. 4b shows that the equilibrium conversion degree of methane increases with an increasing β ratio. However, higher β ratios result in synthetic fuels with higher concentrations of the noncombustible components such as N_2, H_2O and CO_2. The higher concentrations of noncombustible components lead to decreasing the LHV of synthetic fuel. Thus, the optimal β ratio should be βε(1.0; 2.0).

The curves from Fig. 4 were obtained by accomplishment the thermodynamic analysis. The chemical system of thermochemical regenerator (Fig. 2) was considered. Inlet gas mixture is natural gas and flue gases. Outlet gas mixture is synthetic fuel (product gas). The material balance equation and law of mass action was written for that chemical system [33–35].

The equations of material balance of chemical elements for the stationary mode is as follows:

$$\sum_{j=1}^{l} n_{ij} = 0 \quad (10)$$

where n_{ij} is the flow of the i-th component in the j-th input or output, l - total number of inputs and outputs; streams that enter the system are considered positive; flows that divert from the system are considered negative.

The laws of mass action for reactions of steam reforming (4), dry reforming (5) and water-gas shift reaction (6) [36–38]:

$$K_{p(4)} = \frac{p_{H_2} \cdot p_{CO_2}}{p_{CO} \cdot p_{H_2O}}, \quad (11)$$

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$$K_{p(5)} = \frac{\left(p_{H_2}\right)^3 \cdot p_{CO}}{p_{CH_4} \cdot p_{H_2O}}, \tag{12}$$

$$K_{p(6)} = \frac{\left(p_{H_2}\right)^2 \cdot \left(p_{CO}\right)^2}{p_{CH_4} \cdot p_{CO_2}}, \tag{13}$$

Here, p_i is the partial pressure of i-gas.

In the first stage was determined a methane conversion to carbon dioxide (x_{CO2}) in the reaction (6). The system of equations that include the material balance equation for thermochemical regenerator and the law of mass action for reaction (6) were resolved. After that, the equation of law mass action for reaction (4) and (5) were resolved for determination of the methane conversion in these reactions. Finally, for overall reaction reforming of methane by flue gases was determined the conversion degree of methane as follows:

$$x_{CH_4} = \frac{2}{3} x_{CH_4}^{steam} + \frac{1}{3} x_{CH_4}^{dry} \tag{14}$$

Here, $x_{CH_4}^{steam}$ is the conversion degree of methane in reaction (4); $x_{CH_4}^{dry}$ is the conversion degree of methane in reaction (5).

The results of calculation of conversion degree of methane in thermochemical regenerator are shown in Fig. 4.

Table 2 shows the composition of the product gas (synthetic fuel) obtained in the methane reformer at reforming of 1 kmol of CH_4 with the various ratio β and temperature and at pressure p = 1 bar.

The total mass of the initial substances to be reacted in the reactor is $g_{in} = 112$ kg, at the β = 1 and $g_{in} = 208$ kg, at the β = 2. Table 2 also shows the molar mass of the produced substances μ_{mix}, the quantity of the moles of reformate gas n_μ, the heat released at burning 1 kmol of the product gas Q_k, and the total heat, obtained at burning all the produced gases $\sum Q$ and the ratio of this heat to the one, obtained by burning 1 kmol of methane η_q. These values are determined as

$$n_\mu = G_{in} / \mu_{mix} \tag{15}$$

$$Q_k = 283.1CO + 241.8H_2 + 802.3CH_4 KJ/mol. \tag{16}$$

Here, CO, H_2, CH_4 are the gas composition in volume %. The total quantity of heat, obtained from the product reformate gas is determined as

$$\Sigma Q = n_\mu \; Q_k \tag{17}$$

The ratio of heat, distinguished by burning the product gases in relation to the heat of burning the CH_4 to be reacted during the reforming, is determined from:

$$\eta_q = \Sigma Q / 802325. \tag{18}$$

Table 2 makes it clear that the increasing reforming temperature increases the quantity of the heat obtained by the burning of the product gas in comparison to the burning heat of initial methane. The ratio β in the initial mixture influences the value of the obtained combustion heat of the synthetic fuel. The growth of this ratio raises the completeness of the reforming, i.e. increases exergy of synthetic fuel.

However, it can be seen from Table 2, increasing of ratio β (β > 1) increases the content of noncombustible gases. These noncombustible gases decrease the specific low heat value of the synthetic fuel (kJ/mol). Therefore, for scheme that shows in Fig. 2 need to use ratio β ≈ 1.

Heat balance. Fuel saving

This section discusses the heat balance of a forging furnace. In this type of industrial furnace, steel is heated up from an initial temperature (t_1 = 20 °C) to a set temperature (t_2 = 850 °C). Furnace output is L = 1ton/h of heated steel. Fig. 5 shows the heat balance of two forging furnaces that use thermochemical heat regeneration of waste flue gases via two separate schemes: (1) TCR by steam methane reforming (Fig. 1) and (2) TCR by methane reforming using flue gases (Fig. 2).

The heat-balance inputs and outputs are depicted on the left and right sides of the column, respectively. Heat-balance input (Q_{chem}) is the chemical energy of the natural gas that is used in the industrial furnaces depicted earlier (Figss. 1 and 2); $Q_{h.a.}$ Is the hot air heat; $\Delta Q_{chem.}$ Is the enthalpy of the flue gases that is transformed to the chemical energy of the new synthetic fuel; $Q_{h.sf.}$ Is enthalpy of the synthetic fuel. Heat-balance output ($Q_{h.s}$) is the heat required to raise the temperature of the steel; $Q_{st.}$ Is the heat consumed in the production of steam; $Q_{f.g.}$ Is the waste-heat of flue gases that is lost; Q_{other} encompasses all other losses.

The parts of heat-balance input are following:

Table 2 – The equilibrium composition of the gas mixture at different parameters.

T,K	β	Gas composition in volume, %						μ_{mix},	n_μ,	Q_k,	$\sum Q$,	η_q
		H_2	CO	CH_4	H_2O	CO_2	N_2	kg/kmol	kmol	kJ/mol	kJ	
800	1	17,97	4,84	13,96	6,49	7,41	49,34	22,10	5,09	169,17	861,08	1073
	2	14,57	3,21	6,92	9,48	8,81	57,00	23,78	8,80	99,88	878,94	1095
900	1	29,21	12,82	7,01	3,49	3,52	43,96	19,52	5,71	163,18	931,76	1161
	2	23,22	8,96	2,42	6,83	6,06	52,50	21,78	9,55	100,98	964,36	1202
1000	1	36,69	18,05	2,42	1,42	1,00	40,43	17,83	6,21	159,25	988,94	1232
	2	26,72	12,36	0,27	6,20	4,11	50,34	20,81	9,96	101,78	1013,73	1263
1100	1	39,52	19,77	0,78	0,53	0,25	39,16	17,23	6,41	157,80	1011,50	1261
	2	26,57	13,34	0,01	6,70	3,30	50,08	20,68	10,01	102,11	1022,12	1274
1200	1	40,43	20,24	0,28	0,20	0,08	38,78	17,05	6,47	157,32	1017,86	1269
	2	26,07	13,84	0,01	7,20	2,80	50,08	20,67	10,01	102,31	1024,12	1277

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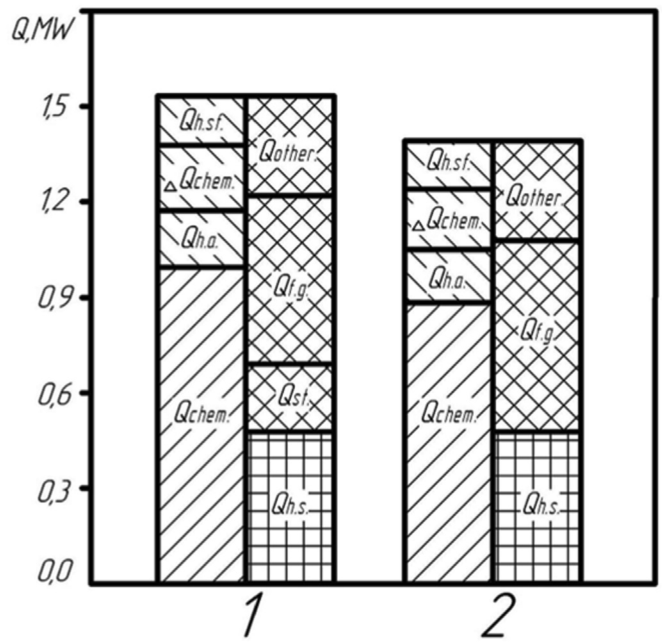

Fig. 5 – The heat balance of the forging furnace (L = 1ton/h) with the thermochemical heat regeneration of waste flue gases.

$$Q_{chem} = B \cdot Q_{lhw}, \tag{19}$$

where B is fuel consumption, m^3 (kg); Q_{lhw} is the LHV, kJ/m^3 (kJ/kg).

$$Q_{h.a.} = V_{h.a.} \cdot t_{h.a.} \cdot c_p, \tag{20}$$

where $V_{h.a.}$ Is required air for combustion, m^3; $t_{h.a.}$ Is temperature of hot air, ^0C; c_p is heat capacity of hot air, $kJ/(m^3 \cdot K)$.

$$Q_{s.f.} = V_{s.f.} \cdot t_{s.f.} \cdot c_p, \tag{21}$$

where $V_{s.f.}$ Is volume of synthetic fuel, m^3; $t_{s.f.}$ Is temperature of syn.fuel, ^0C; c_p is heat capacity of syn.fuel, $kJ/(m^3 \cdot K)$.

$$\Delta Q_{chem} = x_{CH4} \cdot Q, \tag{22}$$

where Q represents the thermal effect of the overall reaction (7) at stoichiometric reagent consumption, $[kJ/m^3]$ and x_{CH4} represents the conversion degree of methane.

The parts of heat-balance output are following:

$$Q_{h.s.} = \frac{100 - \delta}{100} G \cdot (t_2 - t_1) \cdot c_p\ , \tag{23}$$

where δ is waste of metal; G is metal mass, kg; t_2, t_1 is steal temperature in outlet and inlet, respectively, ^0C; c_p is heat capacity of steal, $kJ/(kg \cdot K)$.

$$Q_{f.g.} = V_{f.g.} \cdot t_{hf.g.} \cdot c_p, \tag{24}$$

where $V_{s.f.}$ Is volume of flue gases, m^3; $t_{s.f.}$ Is temperature, ^0C; c_p is heat capacity, $kJ/(m^3 \cdot K)$.

$$Q_{st.} = D_{st.} \cdot h_{st.}, \tag{25}$$

where $D_{st.}$ Is steam consumption, kg; $h_{st.}$ Is enthalpy of steam, kJ/kg.

As can be seen in Fig. 5, the chemical energy of the natural gas ($Q_{chem.}$) for scheme in Fig. 2 is approximately 12% lower than that of scheme in Fig. 1. This is due to the fact that the heat loss for steam production in scheme in Fig. 2 is eliminated, whereas the heat loss for the steam production in scheme in Fig. 1 accounts for approximately 10% of the heat-balance input. In scheme in Fig. 2, the hot air heat is lower than that in scheme in Fig. 1. This is due to the fact that scheme in Fig. 2 has a decreased consumption of natural gas. Thus, though each scheme realizes close to a full heat recovery, scheme in Fig. 2 is more efficient in its consumption of natural gas.

The fuel consumption is calculated for both schemes of the forging furnaces and is compared above (Figss. 1 and 2). Additionally, the fuel consumption calculation for a scheme that uses thermal regeneration by heating air to 500 °C has been included (3).

The fuel consumption can be calculated by following equation:

$$b = \frac{Q_{h.s.}}{G \cdot \eta}, \tag{26}$$

where b is specific fuel consumption, kg/ton; η is the coefficient of efficiency. Here, η can be determined from Fig. 4 as follows:

$$\eta = 1 - \sum q_{loss}, \tag{27}$$

where q_{loss} is a heat loss with flue gases, with water cooling, etc.

Fig. 6 shows the fuel consumption in the process that involves heating steel with the forging furnace from 20 °C to 850 °C.

As can be seen from Fig. 6, fuel consumption is the lowest in the case of the forging furnace that employs TCR by reforming methane using flue gases. Thus, this method of recovering heat from waste flue gases offers the most advanced option. The results show that the fuel consumption

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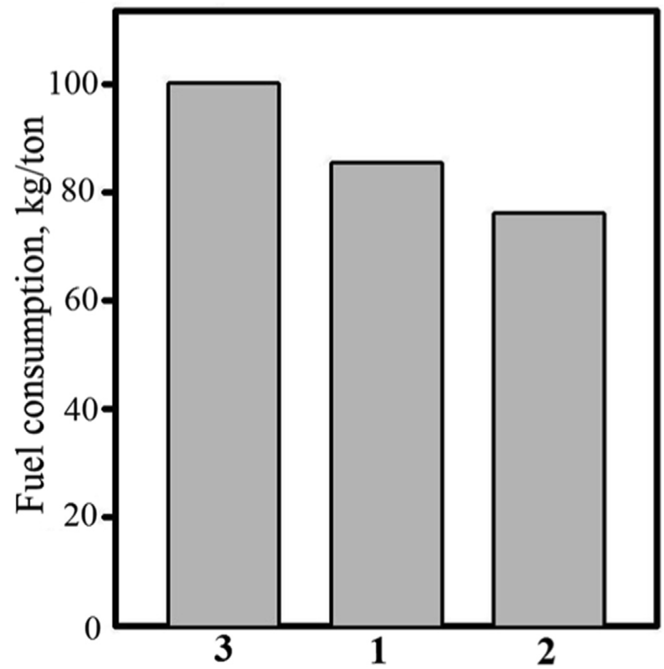

Fig. 6 — The coal equivalent consumption (Q = 29.3 MJ/kg) for the forging furnace.

of TCR with methane reforming using flue gases is 23% lower than that of the conventional thermal regeneration and 16% lower than that of TCR by SMR. Thus, if energetic plants carry out TCR by reforming methane using flue gases, high energy efficiency can be achieved and, as a result, fuel consumption can be lowered considerably.

The results of this study can be used for calculation of economic efficiency of using the thermochemical waste-heat recuperation systems in the real forging furnaces. The fuel consumption in the forging furnace with TCR by methane reforming with combustion products is about 80 kg of fuel per 1 ton of steel (LHV of fuel is 29.3 MJ/kg). The heat for steel heating can be considered as an exergy in that system. Thus, the product of the fuel mass flow rate and the calorific value indicates the exergy in such a system [39,40].

Conclusion

This paper has proposed an innovative method of regeneration technology for application in industrial furnaces in which heat recovery is carried out using methane reforming using flue gases. This method of heat recovery was compared to thermochemical heat regeneration by SMR. A big disadvantage of TCR by SMR is that methane reforming requires steam production. This disadvantage can be countered by the use of thermochemical waste-heat regeneration by reforming methane using flue gases. If in the conventional industrial furnaces conversion of chemical energy of fuel into heat energy occurs by fuel combustion, then in the industrial furnaces with TCR the conversion of chemical energy of fuel is divided into two stages. The first stage acts to increase the LHV of the initial natural gas by creating a new synthetic fuel with a LHV greater than that of the initial natural gas. The second stage involves the combustion of this new synthetic gas.

The thermodynamic aspects of the proposed technology were also considered. It has been shown that the enthalpy of the flue gases are transformed into chemical energy of the new synthetic fuel is directly proportional to the conversion degree of methane. The variations of conversion degree of methane concerning temperature and β ratio were calculated. The heat balances of both the compared method of TCR by methane reforming using flue gases and TCR by SMR were calculated using a process simulator. The results show that the fuel consumption of the TCR by reforming methane using flue gases is drastically reduced by 23% and 16% in comparison to that of the conventional thermal regeneration and TCR by SMR, respectively. In conclusion, the improved energy efficiency and the reduced fuel consumption realized by the proposed method of heat regeneration – thermochemical regeneration by reforming methane using flue gases – offers a very promising technology for saving energy and reducing negative effects on the environment.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

η – Energy efficiency (coefficient of efficiency), [–]
P – Economy of fuel (natural gas) in furnace, [%]
H – Enthalpy of flue gases, [kJ/m^3]
r – Regeneration rate, [–]
t – Temperature, [^{0}C]
T – Temperature, [K]
t – Temperature, [^{0}C]
p – Pressure, [bar]
g_{in} – Total mass of the initial substances, [kg]
Q_k – specific low heat value, [kJ/mol]
$\sum Q$ – total low heat value, [kJ]
μ – molar mass, kg/kmol
n_μ – quantity of the moles, [kmol]
η_q – coefficient of heat transformation, [–]
b – Specific fuel consumption, [kg/ton]
B – Consumption of natural gas, [m^3]
c_p – Heat capacity, [kJ/(m^3·K)]
D_{st} – Steam consumption, [kg]
h – Enthalpy, [kJ/kg]
G – Mass of metal, [kg]
q_{loss} – heat loss, [–]
K_p – equilibrium constant, [–]
x_{CH4} – Conversion degree of methane, [–]
x_{CO2} – Conversion degree of methane to carbon dioxide, [–]
β – Ratio ($H_2O + CO_2$)/CH_4
L – furnace output, [t/h]
k, a, b, c, d, e – stoichiometric coefficients for reforming reactions
SMR – Steam methane reforming
TCR – Thermochemical regeneration
LHV – Low heat value
TR – Thermochemical regenerator

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